Introduction
Kummer lines
Doubling
Diff. add.
Classification
Conclusion

Arithmétique des droites de Kummer

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Sous la direction de Razvan Barbulescu et Damien Robert

9 juillet 2025

Introduction : cryptographie à clé publique

Échange sécurisé de messages (1/2)

Alice —————————————— Bob

Échange sécurisé de messages (1/2)

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Échange sécurisé de messages (2/2)

Alice et Bob partagent une clé secrète 🔑.

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Échange sécurisé de messages (2/2)

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Échange sécurisé de messages (2/2)

Alice et Bob partagent une clé secrète 🔑.

Comment faire en sorte qu'Alice et Bob aient la même clé secrète 🔑 ?

Groupes abéliens

Groupe abélien $(G, +)$, $g, h, k \in G$

- $g + h \in G$;
- $g + h = h + g$;
- $(g + h) + k = g + (h + k)$;
- $g + 0_G = g$;
- $g + (-g) = 0_G$.

Exemples

- Entiers $\mathbb{Z}$: $\ldots, -3, -2, -1, 0, 1, 2, 3, \ldots$;
- Restes de la division euclidienne $\mathbb{Z}/24\mathbb{Z} = \{\overline{0}, \overline{1}, \ldots, \overline{23}\}$: $\overline{23} + \overline{8} = \overline{7}$.

Notation : si $n \in \mathbb{N}$, $g \in G$, $n \cdot g = g + g + \cdots + g$.

Échange de clés de Diffie-Hellman

- On représente les messages par des éléments d'un groupe public cyclique.
- Alice et Bob choisissent un nombre secret chacun (a et b respectivement).

Groupe public : $G = \langle g \rangle$

Alice	Bob
Secret a	Secret b

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$$a \cdot g_b = (ab) \cdot g \quad \text{🔑} \quad b \cdot g_a = (ab) \cdot g$$

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Problème computationnel de Diffie-Hellman (CDHP)

En connaissant g, g_a et g_b, peut-on calculer la clé secrète 🔑 ?

Logartihme discret

Problème computationnel de Diffie-Hellman (CDHP)

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Problème du logarithme discret (DLP)

En connaissant g et h tels que $a \cdot g = h$, peut-on calculer a ?

- Exemple sur $G = (\mathbb{Z}, +)$: si $ag = h$, alors $a = h/g$.
- Exemple sur $G = (\mathbb{Z}/p\mathbb{Z}, +)$: si $ag \equiv h \bmod p$, alors $a \equiv g^{-1}h \bmod p$.
- Exemple sur $G = (\mathbb{F}_p^*, \times)$: si $g^a \equiv h \bmod p$, $a \equiv$??? $\bmod p - 1$.

Logartihme discret

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Connaît-on d'autres groupes où l'on sait calculer facilement ?

Courbes elliptiques (carac $k \neq 2, 3$)

- Weierstrass court (cas général) :

$$E : y^2 = x^3 + ax + b$$

- Courbes de Montgomery :

$$E : By^2 = x(x^2 + \mathcal{A}x + 1)$$

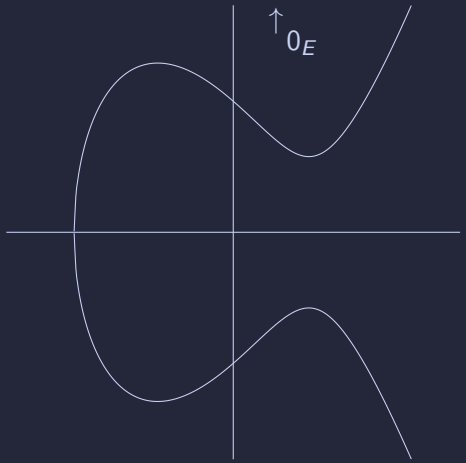

Figure : Points réels de $E : y^2 = x^3 - 3x + 3$

Loi de groupe : définition

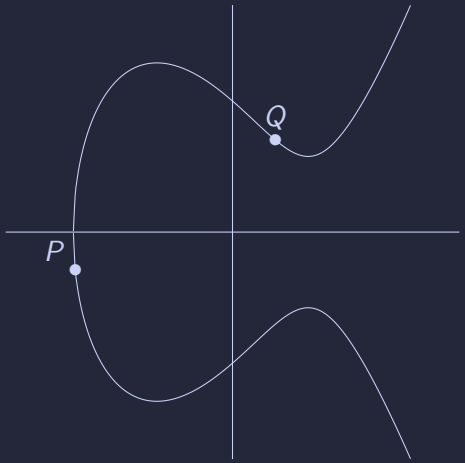

Figure : Addition de $P \neq Q$

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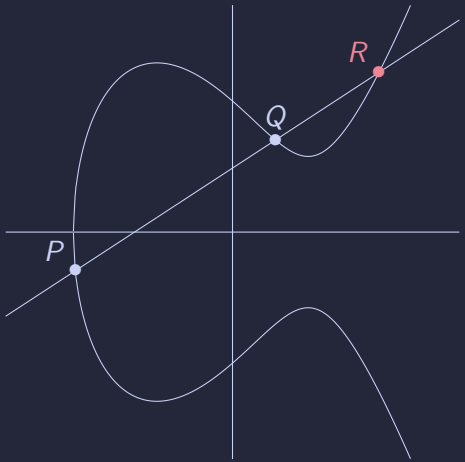

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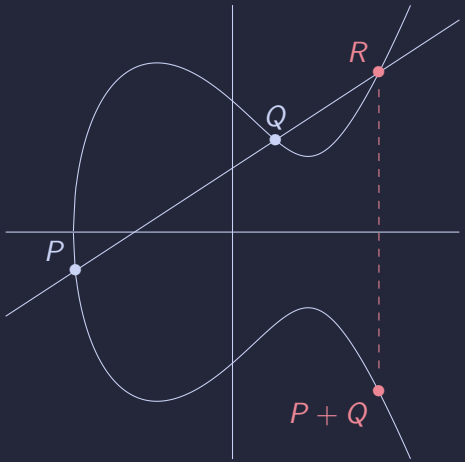

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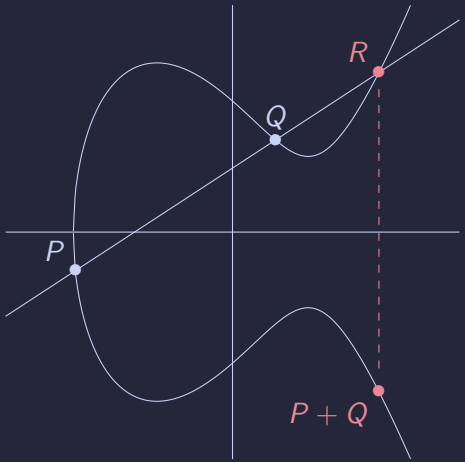

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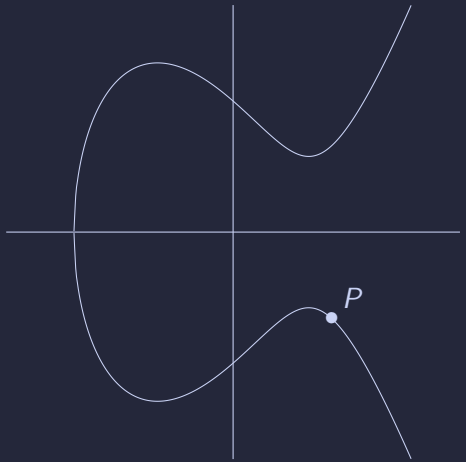

Figure : Addition de P et P

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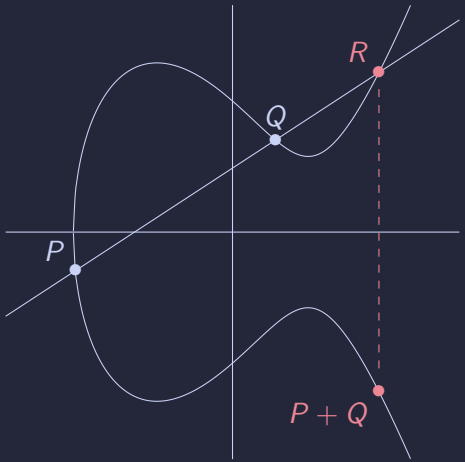

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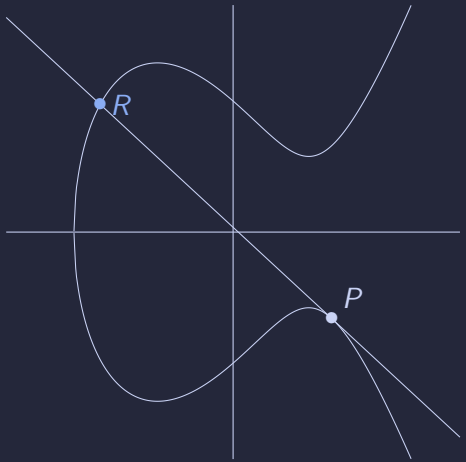

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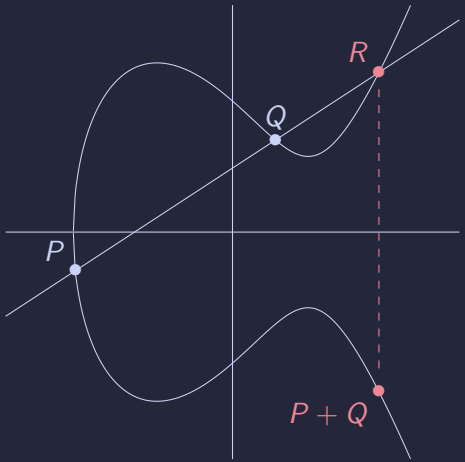

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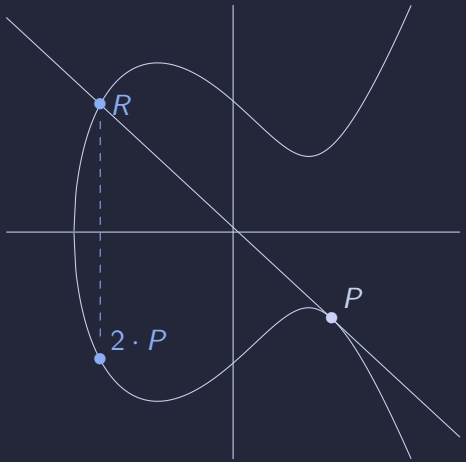

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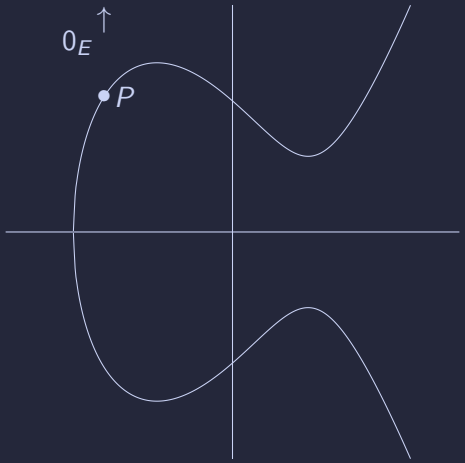
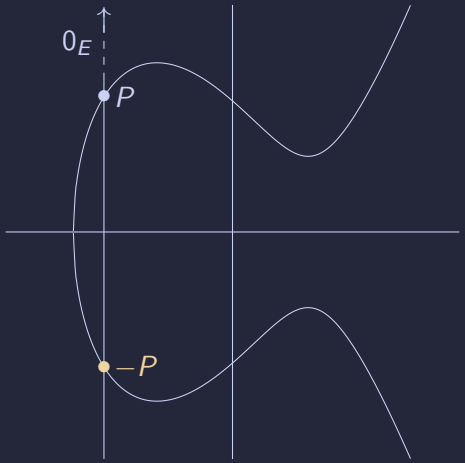


Figure : Élément neutre et opposé de P

Loi de groupe : propriétés



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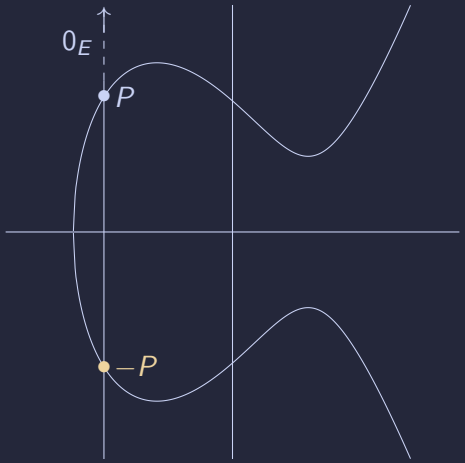

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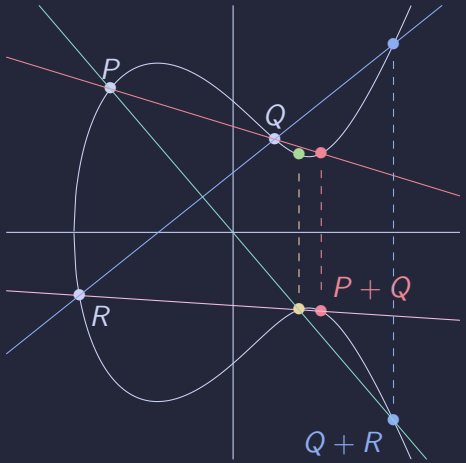

Figure : $(P+Q)+R = P+(Q+R)$

Différentes équations

$$y_W^2 = x_W^3 + px_W + q$$

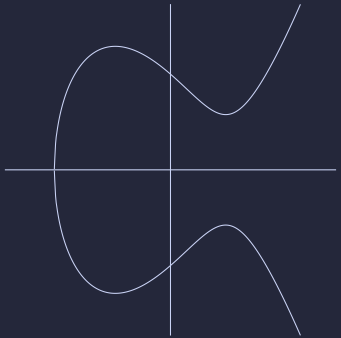

Figure : Weierstrass court

$$By_M^2 = x_M(x_M^2 + \mathcal{A}x_M + 1)$$

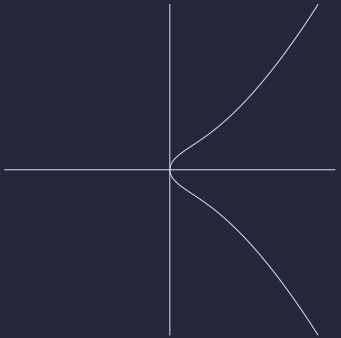

Figure : Montgomery

$$ax_E^2 + y_E^2 = 1 + dx_E^2y_E^2$$

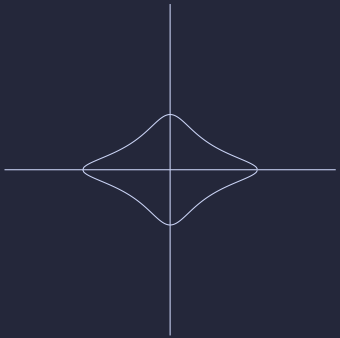

Figure : Edwards tordue

Problématique

Étant donné une courbe elliptique E et des points $P, Q \in E$:

Question 1

Comment trouver des formules pour $P + Q$?

Question 2

Comment calculer efficacement $n \cdot P$?

Question 3

Quelles courbes ont une arithmétique rapide ?

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Arithmetic of Kummer lines

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Outline

1. Introduction : cryptographie à clé publique
2. Models of Kummer lines
   - Montgomery ladder
   - Definition
3. Arithmetic: Doubling
   - Isogenies
   - Doubling formulas
   - Hybrid ladder
4. Arithmetic: Differential addition
   - Half differential addition formulas
   - Half ladder
5. Classification

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Models of Kummer lines

x-only arithmetic

From $x(P), y(P), x(Q), y(Q)$, we can compute $x(P+Q)$ and $y(P+Q)$.
What if we only keep $x(P)$ and $x(Q)$?

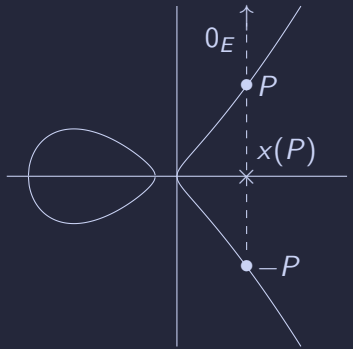

Figure: A Montgomery curve

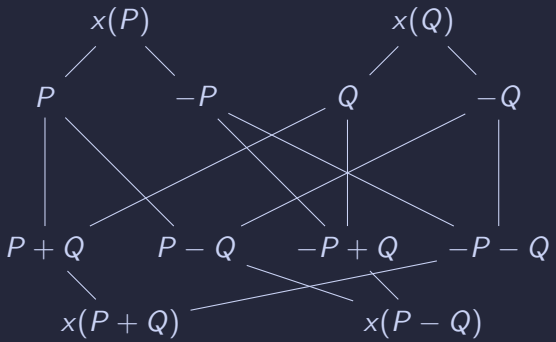

Figure: Two possible choices

x-only arithmetic

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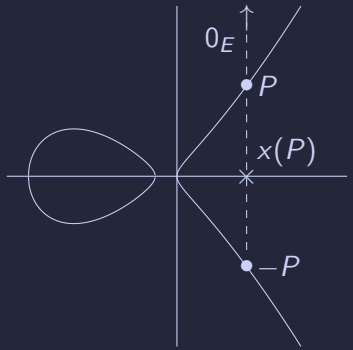

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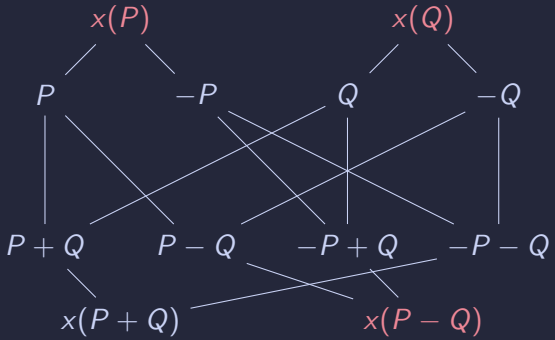

Figure: Two possible choices

If we know $x(P)$, $x(Q)$, $x(P-Q)$, we can compute $x(P+Q)$.

Arithmetic on $y^2 = x(x^2 + \mathcal{A}x + 1)$[1]

Differential addition ($3M + 2S$)

$u := (X_P + Z_P)(X_Q - Z_Q)$, $v := (X_P - Z_P)(X_Q + Z_Q)$.

$$X_{P+Q} = (u+v)^2, \quad Z_{P+Q} = \frac{X_{P-Q}}{Z_{P-Q}}(u-v)^2.$$

Doubling ($2M + 2S + 1m_0$, $d = \frac{\mathcal{A}+2}{4}$)

$u := (X_P + Z_P)^2$, $v := (X_P - Z_P)^2$, $t := u - v$.

$$X_{2.P} = uv, \quad Z_{2.P} = t(v + dt).$$

[1] P. L. Montgomery. 'Speeding the Pollard and elliptic curve methods of factorization'. In: *Mathematics of Computation* 48 (1987), pp. 243–264.

Algorithm 1: Montgomery ladder step

Input: $R = m \cdot P$, $S = (m+1) \cdot P$, b a bit
Output: $(2 \cdot R, R + S)$ if $b = 0$ $(R + S, 2 \cdot S)$ if $b = 1$
Data: The point P

```
Function xDBLADD(R, S, b):
    if b = 0 then
        S ← DiffAdd(R, S, P);
        R ← Doubling(R);
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$n = 11 = \overline{1011}^2$

0_E ——— P

Figure: Montgomery ladder

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$n = 11 = \overline{1011}^2$

1

$$\begin{tikzcd} 0_E \arrow[r, dash] \arrow[d] & P \arrow[d] \arrow[dl] \\ P \arrow[r, dash] & 2P \end{tikzcd}$$

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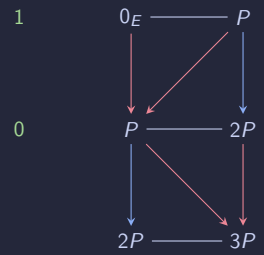

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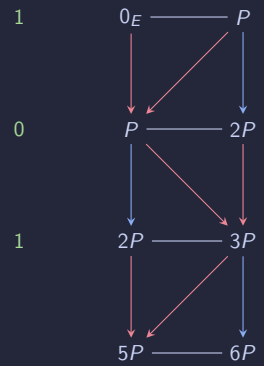

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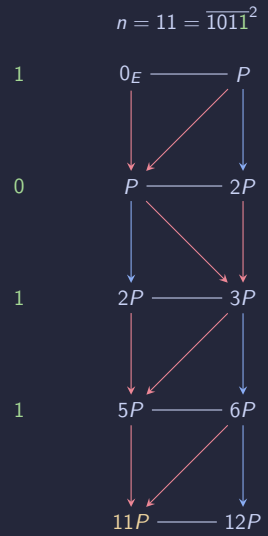

Figure: Montgomery ladder

Kummer line model of a Montgomery curve

$$E : BY^2Z = X(X^2 + \mathcal{A}XZ + Z^2)$$

If $P = (X : Y : Z)$, then $-P = (X : -Y : Z)$;
$0_E = (0 : 1 : 0)$.

Montgomery XZ-coordinates

$$x : E \to \mathbb{P}^1, \begin{cases} 0_E & \mapsto \infty := (1 : 0), \\ (X : Y : Z) & \mapsto \frac{X}{Z} := (X : Z). \end{cases}$$

Since $x(P) = x(-P)$:

- $\#x^{-1}(x(P)) = 1$ when $P = -P$;
- $\#x^{-1}(x(P)) = 2$ otherwise.

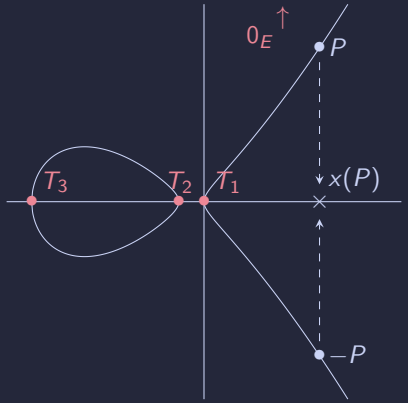

Figure: Ramification on the curve

Kummer line model of a Montgomery curve

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If $P = (X : Y : Z)$, then $-P = (X : -Y : Z)$;
$0_E = (0 : 1 : 0)$.

Montgomery XZ-coordinates

$$x : E \to \mathbb{P}^1, \begin{cases} 0_E & \mapsto \infty := (1 : 0), \\ (X : Y : Z) & \mapsto \frac{X}{Z} := (X : Z). \end{cases}$$

Since $x(P) = x(-P)$:

- $\#x^{-1}(x(P)) = 1$ when $P = -P$;
- $\#x^{-1}(x(P)) = 2$ otherwise.

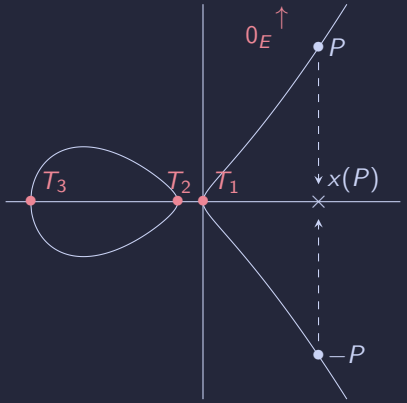

Figure: Ramification on the curve

Can we reproduce this approach?

Kummer line: general definition

Definition

A Kummer line of an elliptic curve E is:

- A degree 2 covering $\pi : E \to \mathbb{P}^1 \simeq E/\pm 1$:

$$\pi^{-1}(\pi(P)) = \{-P, P\}.$$

- 4 ramification points, which correspond to the 2-torsion:

$$\pi^{-1}(\pi(T)) = \{T\} \text{ for } T \in E[2].$$

A map between Kummer lines $\varphi : \mathbb{P}^1 \to \mathbb{P}^1$ maps the ramification points to ramification points.

Examples

Montgomery model $M(a:b)$: $y^2 = x(x - a/b)(x - b/a)$, $a/b \in \overline{k}$

$$\pi : (x, y) \mapsto (x : 1),\ 0_E \mapsto (1 : 0).$$

$$0_E = (1 : 0)^*, \quad T_1 = (0 : 1), \quad T_2 = (a : b), \quad T_3 = (b : a).$$

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Theta squared model $\theta_s(a:b)$: $y^2 = x(x-a/b)(x-b/a)$, $a/b \in k$

$$\pi : (X:Y:Z) \mapsto (aX - bZ : bX - aZ),\ 0_E \mapsto (a:b).$$

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Theta model $\theta(A:B)$: $y^2 = x(x - a/b)(x - b/a)$, $a/b \in k$

$$(A^2 : B^2) = (a+b : a-b), \text{ assume } A/B \in k.$$

$$\pi : (X:Y:Z) \mapsto (A(X-Z) : B(X+Z)), \; 0_E \mapsto (A:B).$$

$$0_E = (A:B)^*, \quad T_1 = (-A:B), \quad T_2 = (B:A), \quad T_3 = (-B:A).$$

Arithmetic on $\theta_s(a:b)$[2]

Differential addition ($3M + 2S + 1m_0$, Montgomery: $3M + 2S$)

$u := (X_P + Z_P)(X_Q + Z_Q)$, $v := \frac{a+b}{a-b}(X_P - Z_P)(X_Q - Z_Q)$.

$$X_{P+Q} = (u+v)^2, \quad Z_{P+Q} = \frac{X_{P-Q}}{Z_{P-Q}}(u-v)^2.$$

Doubling ($4S + 2m_0$, Montgomery: $2M + 2S + 1m_0$)

$u := (X_P + Z_P)$, $v := \frac{a+b}{a-b}(X_P - Z_P)$.

$$X_{2.P} = (u+v)^2, \quad Z_{2.P} = \frac{a}{b}(u-v)^2.$$

[2] P. Gaudry and D. Lubicz. 'The arithmetic of characteristic 2 Kummer surfaces and of elliptic Kummer lines'. In: *Finite Fields Their Appl.* 15.2 (2009), pp. 246–260.

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Arithmetic: Doubling

Isogenies (between elliptic curves): definition

Definition

Surjective homomorphism $f : E \to E'$ with finite kernel.
$\rightsquigarrow$ Induces a field extension $f^* : k(E') \to k(E)$: separable, $\deg f$, $\deg_s f$, ...

Some examples:

- the multiplication by $m \in \mathbb{Z}$: $[m]_E : P \in E \mapsto m \cdot P$;
- $f : (E : y^2 = x(x^2 + 3x + 1)) \to (E' : y^2 = (x-2)(x+2)(x+3))$ where

$$f : (x, y) \mapsto \left(\frac{x^2+1}{x}, y\frac{x^2-1}{x^2} \right).$$

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At the Kummer line level, with the x-coordinate model, the induced map is

$$\varphi : (X : Z) \mapsto (X^2 + Z^2 : XZ).$$

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Isogenies: cryptography

Why are we interested in these objects?

- Natural to study the morphisms of elliptic curves;
- More importantly, post-quantum cryptography candidates.

$$\begin{tikzcd} E \arrow[r, "f_A"] \arrow[d, "f_B"'] & E_A \arrow[d, "f_{AB}"] \\ E_B \arrow[r, "f_{BA}"'] & E_{AB} \end{tikzcd}$$

Figure: SIDH protocol (attacked in 2022)

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Figure: SIDH protocol (attacked in 2022)

$$\begin{tikzcd} E_2^2 \arrow[r, "\alpha_2"] & E_2^2 \\ E_1^2 \arrow[u, "\Sigma"] \arrow[r, "\alpha_1"'] & E_1^2 \arrow[u, "\Sigma"'] \end{tikzcd}$$

$$\rightsquigarrow F = \begin{pmatrix} \alpha_1 & \widetilde{\Sigma} \\ -\Sigma & \alpha_2 \end{pmatrix} \in \mathrm{End}(E_1^2 \times E_2^2)$$

Figure: SQISignHD protocol

Isogenies: dual

When f is separable, the kernel fully describes f and $\#\ker f = \deg f$.
A d-isogeny is a (separable) isogeny with cyclic kernel of order d.

Theorem

If $f : E \to E'$ is an isogeny between elliptic curves, then there is an isogeny $\widetilde{f} : E' \to E$, called the dual and such that

$$\widetilde{f} \circ f = [\deg f]_E \quad \text{and} \quad f \circ \widetilde{f} = [\deg f]_{E'}.$$

In particular, with a 2-isogeny f, we can factor the multiplication by 2, giving fast doubling formulas.

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Goal

Are there fast 2-isogeny formulas between models of Kummer lines?

A methodology to find 2-isogenies formulas[3]

If $f : E \to E'$ is a 2-isogeny and $\ker f = \{0_E, T\}$, then $f(\cdot + T) = f$.
At the Kummer line level $\pi : E \to \mathbb{P}^1$:

1. T is of 2-torsion, hence the translation by T descends to $\mathbb{P}^1$: $t_T : \mathbb{P}^1 \to \mathbb{P}^1$.
2. It is a homography (isomorphism of $\mathbb{P}^1$), and can be computed from the ramification points of π.

[3] D. Robert and N. S. 'Computing 2-isogenies between Kummer lines'. In: *Communications in Cryptology* 1.1 (2024), p. 26

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3. t_T acts on X^2, Z^2, XZ: helps to build invariants u, v of degree 2 by t_T:

$$\varphi : (X : Z) \mapsto (u(X : Z) : v(X : Z)), \text{ such that } \varphi \circ t_T = \varphi.$$

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4. We can derive the ramification on the right model from the ramification of π, and shape it with a homography.

[3] D. Robert and N. S. 'Computing 2-isogenies between Kummer lines'. In: *Communications in Cryptology* 1.1 (2024), p. 26

Some old formulas, in a unified manner (1/2)

Starting from $M(a:b)$ associated to $y^2 = x(x-a/b)(x-b/a)$ with ramification

$$0_E = (1:0)^*, \quad T_1 = (0:1), \quad T_2 = (a:b), \quad T_3 = (b:a).$$

Kernel T_1 (Hisil, Renes, 2019)

Assume $a/b \in k$, set $(a':b') = (a+b : a-b)$. Then $\varphi : M(a:b) \to \theta_s(b':a')$:

$$\varphi : (X:Z) \mapsto \left(b'(X+Z)^2 : a'(X-Z)^2\right).$$

The dual is $\widetilde{\varphi} : \theta_s(b':a') \to \theta_s(a:b)$:

$$\widetilde{\varphi} : (X:Z) \mapsto \left(b(X+Z)^2 : a(X-Z)^2\right).$$

Some old formulas, in a unified manner (2/2)

$$0_E = (1:0)^*, \quad T_1 = (0:1), \quad T_2 = (a:b), \quad T_3 = (b:a).$$

Kernel T_2 (Renes, 2018, $T_2' = (A:B)$ rational above T_2)

Set $(A':B') = (A+B : A-B)$, then $\varphi : M(a:b) \to \theta_s(A^2 : B^2)$:

$$\varphi : (X:Z) \mapsto \left((B'(X+Z) + A'(X-Z))^2 : (B'(X+Z) - A'(X-Z))^2\right).$$

Kernel T_1 (de Feo, Jao, Plut, 2014, $\widetilde{T_1} = (r:s)$ rational above $T_1' = (1:1)$)

Set $(\gamma:\delta) = (4rs : (r-s)^2)$, then $\varphi : M(a:b) \to M(-\gamma : 4\delta)$:

$$\varphi : (X:Z) \mapsto \left(\gamma(X-Z)^2 : 4\delta XZ\right).$$

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Even more formulas, some of which are new: Legendre to Montgomery, theta to Montgomery, theta to theta, ...

Doubling up to a 2-torsion point

- we have an isomorphism $\psi : \theta_s(a : b) \to M(a : b),\ P_\theta \mapsto P_M$;
- the map $\overline{\psi} : \theta_s(a : b) \to M(a : b),\ (X : Z) \mapsto (X : Z)$ is a bijection, sending P_θ to $P_M + T$ where T corresponds to the ramification point $(a : b)$.

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$$\begin{tikzcd} M(a:b) \arrow[r, "\varphi"] & \theta_s(b':a') \arrow[r, "\widetilde{\varphi}"] & \theta_s(a:b) \arrow[r, "\overline{\psi}"] & M(a:b) \\ P_M \arrow[rrr, mapsto] & & & 2 \cdot P_M + T \end{tikzcd}$$

Figure: Doubling up to a 2-torsion point

$\texttt{DoublingTranslation}(P) := 2 \cdot P + T$ ($4S + 2m_0$ compared to $2M + 2S + 1m_0$)

Algorithm 2: Hybrid ladder step

Input: $R, S \in \{P, P+T\}$, b a bit

Output: $(2 \cdot R + T, R + S)$ if $b = 0$
$(R + S, 2 \cdot S + T)$ if $b = 1$

Data: The points P and $P + T$

```
Function xDBLADD(R, S, b):
    Q ← R − S;                          // pre-computed
    if b = 0 then
        S ← DiffAdd(R, S, Q);
        R ← DoublingTranslation(R);
    else if b = 1 then
        R ← DiffAdd(R, S, Q);
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    end
    return (R, S);
```

$n = 11 = \overline{1011}^2$

0_E ——— P

Figure: Hybrid ladder

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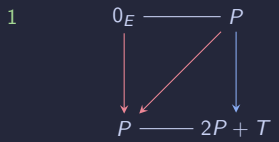

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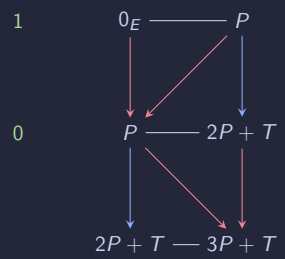

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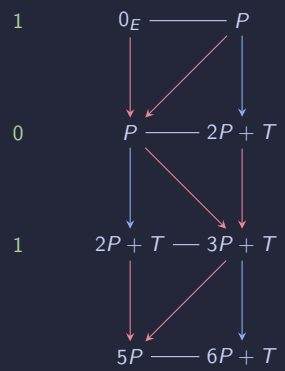

Figure: Hybrid ladder

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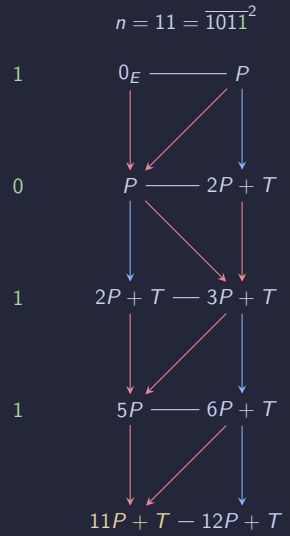

Figure: Hybrid ladder

Hybrid ladder: computational cost

- differential addition on a Montgomery curve: $2M + 2S + 1m$;
- differential addition in theta coordinates: $2M + 2S + 1m + 1m_0$;
- doubling on a Montgomery curve: $2M + 2S + 1m_0$;
- doubling up to a 2-torsion point: $4S + 2m_0$.

Variant	Cost per bit
Montgomery ladder	$4M + 4S + 1m + 1m_0$
Theta ladder	$2M + 6S + 1m + 3m_0$
Hybrid ladder, our contribution	$2M + 6S + 1m + 2m_0$

Table: Comparison of different variants

When $S = 0.66M$, we need $1m_0 \leq 0.66M$ for the hybrid ladder to be faster than the Montgomery ladder.

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Arithmetic: Differential addition

The differential addition isogeny

Starting point

Differential addition formulas come from an isogeny on $E \times E$!

$$\begin{aligned} F : E \times E &\to E \times E \\ (P, Q) &\mapsto (P + Q, P - Q) \end{aligned}$$

The differential addition isogeny

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$$F : E \times E \to E \times E$$
$$(P, Q) \mapsto (P + Q, P - Q)$$

$$\begin{tikzcd}
E \arrow[dr, "\varphi"] \arrow[dd, "{[2]_E}"'] & \\
& E' \arrow[dl, "\widetilde{\varphi}"] \\
E &
\end{tikzcd}$$

Figure: Factoring doubling

The differential addition isogeny

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$$F : E \times E \to E \times E$$
$$(P, Q) \mapsto (P + Q, P - Q)$$

$$\Phi : E \times E \to E' \times E'$$
$$(P, Q) \mapsto (\varphi(P), \varphi(Q))$$

$$\begin{tikzcd} E \arrow[dd, "{[2]_E}"'] \arrow[dr, "\varphi"] & \\ & E' \arrow[dl, "\widetilde{\varphi}"] \\ E & \end{tikzcd}$$

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$$\begin{tikzcd} E \times E \arrow[r, "\Phi"] \arrow[d, "F"'] & E' \times E' \arrow[dl, dashed, "?"] \\ E \times E & \end{tikzcd}$$

Figure: Factoring diff. add. ?

$$\begin{tikzcd} E \arrow[dr, "\varphi"] \arrow[dd, "{[2]_E}"'] & \\ & E' \arrow[dl, "\widetilde{\varphi}"] \\ E & \end{tikzcd}$$

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$$\begin{tikzcd} E \times E \arrow[r, "\Phi"] \arrow[d, "F"'] & E' \times E' \arrow[d, dashed] \\ E \times E \arrow[r, dashed] & A \end{tikzcd}$$

Figure: Factoring diff. add. ? Not as easy

$$\begin{tikzcd} E \arrow[dr, "\varphi"] \arrow[dd, "{[2]_E}"'] & \\ & E' \arrow[dl, "\widetilde{\varphi}"] \\ E & \end{tikzcd}$$

Figure: Factoring doubling

Half differential addition

$$F : (P, Q) \mapsto (P + Q, P - Q), \quad \Phi : (P, Q) \mapsto (\varphi(P), \varphi(Q)).$$

Definition

Half differential addition formulas relative to φ are formulas such that given $\varphi(P)$, $\varphi(Q)$ and $P - Q$, can compute $P + Q$ on the Kummer line.

Notation

- $P + Q = \texttt{HalfDiffAdd}_\varphi(\varphi(P), \varphi(Q), P - Q)$;
- For consistency, $2 \cdot P = \texttt{HalfDouble}_\varphi(\varphi(P))$ $(= \widetilde{\varphi}(\varphi(P)))$.

Example on $\theta(a : b)$[4]

$$0_E = (a : b)^*, \quad T_1 = (-a : b), \quad T_2 = (b : a), \quad T_3 = (-b : a).$$

Set $(A^2 : B^2) := (a^2 + b^2 : a^2 - b^2)$, assume $A/B \in k$. The 2-isogeny considered is:

$$\varphi : (X : Z) \in \theta(a : b) \mapsto (B(X^2 + Z^2) : A(X^2 - Z^2)) \in \theta(A : B)$$

[4] D. Robert and N. S. 'Halving Differential Additions on Kummer Lines'. In: *EUROCRYPT 2025, Part VI*. vol. 15606, pp. 416–445

Example on $\theta(a:b)$[4]

$$0_E = (a:b)^*, \quad T_1 = (-a:b), \quad T_2 = (b:a), \quad T_3 = (-b:a).$$

Set $(A^2 : B^2) := (a^2 + b^2 : a^2 - b^2)$, assume $A/B \in k$. The 2-isogeny considered is:

$$\varphi : (X:Z) \in \theta(a:b) \mapsto (B(X^2+Z^2) : A(X^2-Z^2)) \in \theta(A:B)$$

$\texttt{HalfDiffAdd}_\varphi(\varphi(P), \varphi(Q), P-Q)$ $(4M)$

$$(X_{P+Q}X_{P-Q} : Z_{P+Q}Z_{P-Q}) = \begin{pmatrix} X_{\varphi(P)}X_{\varphi(Q)} + Z_{\varphi(P)}Z_{\varphi(Q)} \\ X_{\varphi(P)}X_{\varphi(Q)} - Z_{\varphi(P)}Z_{\varphi(Q)} \end{pmatrix}$$

A full differential addition in $\theta(a:b)$ is $3M + 4S + 1m_0$.

[4] D. Robert and N. S. 'Halving Differential Additions on Kummer Lines'. In: *EUROCRYPT 2025, Part VI*. vol. 15606, pp. 416–445

Introduction
Kummer lines
Doubling
Diff. add.
Half differential addition formulas
Half ladder
Classification
Conclusion

Half ladder (1/2)

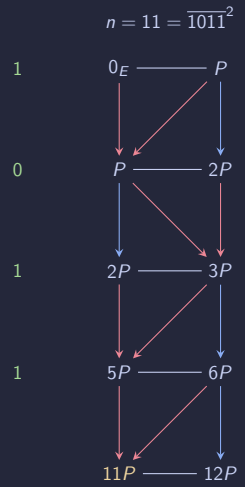
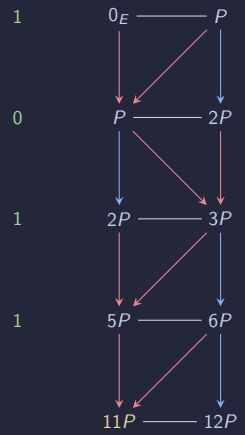


Figure: Montgomery ladder

Half ladder (1/2)

$n = 11 = \overline{1011}^2$



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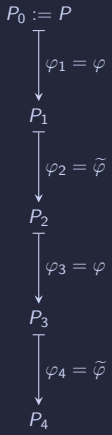

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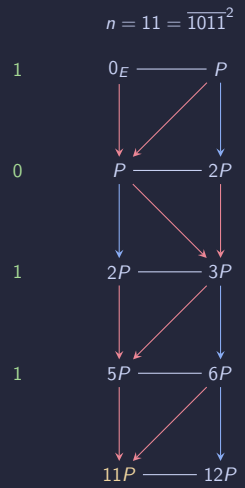

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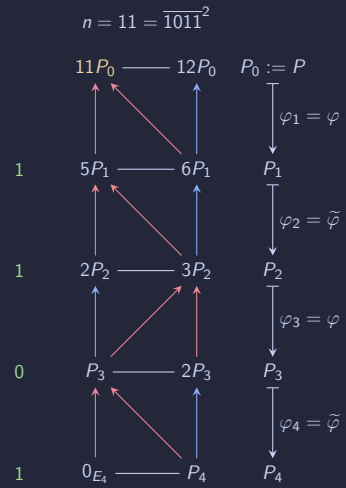

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Algorithm 3: Half ladder step for a 2-isogeny φ

Input: $\varphi(R)$, $\varphi(S)$ where $R = m \cdot P$, $S = (m+1) \cdot P$, b a bit

Output: $(2 \cdot R, R + S)$ if $b = 0$
$(R + S, 2 \cdot S)$ if $b = 1$

Data: The point P

```
Function HalfxDBLADD_φ(φ(R), φ(S), b):
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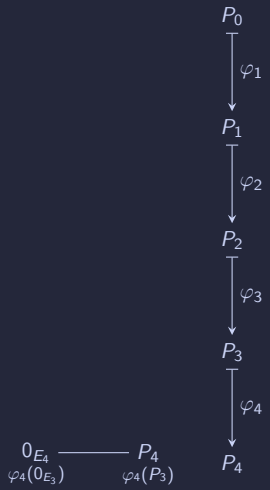

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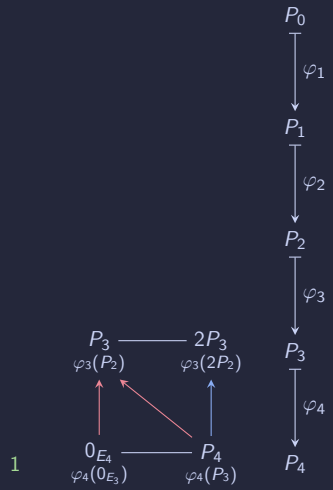

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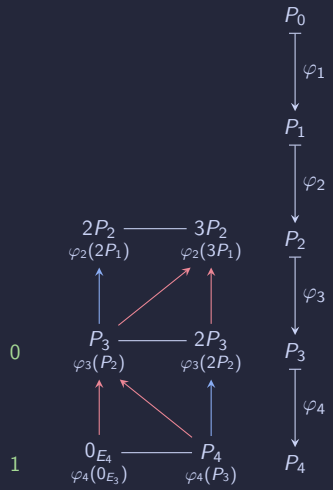

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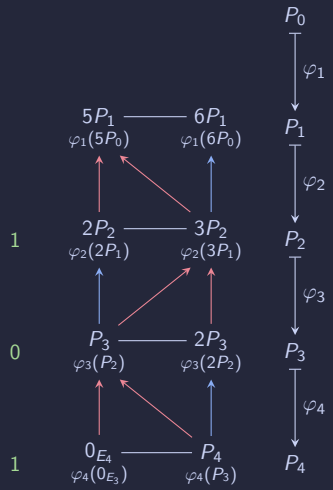

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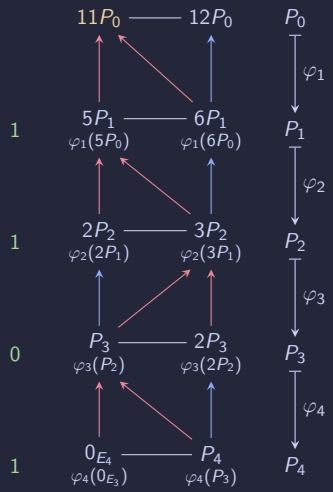

Figure: Half ladder

Computational cost: without pre-computation

On our theta model $\theta(a:b)$ previously studied, with $\varphi_{2i+1} = \varphi$ and $\varphi_{2i} = \widetilde{\varphi}$:

- $\varphi : \theta(a:b) \to \theta(A:B)$ and $\widetilde{\varphi}$: $2S + 1m_0$;
- $\mathtt{HalfDiffAdd}_{\varphi}$ and $\mathtt{HalfDiffAdd}_{\widetilde{\varphi}}$: $4M$;
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	Montgomery ladder	Half ladder, our contribution
Non-normalized base point	$6M + 4S + 1m_0$	$4M + 4S + 2m_0$
Normalized base point	$5M + 4S + 1m_0$ (or $4M + 4S + 2m_0$)	

Table: Ladder cost per bit with no pre-computation

Computational cost: with pre-computation

Algorithm	Pre-computation	Step
Montgomery ladder RtL[5]	$2M + 2S + 1m_0$	$4M + 2S$
Half ladder, our contribution	$2S + 1m_0$	$4M + 2S + 1m_0$

Table: Ladder costs per bit with a pre-computation but no normalization

[5] T. Oliveira et al. 'How to (Pre-)Compute a Ladder - Improving the Performance of X25519 and X448'. In: *SAC 2017*. Vol. 10719. Lecture Notes in Computer Science. Aug. 2017, pp. 172–191

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Table: Ladder costs per bit with a pre-computation but no normalization

Also holds on a Montgomery curve with full 2-torsion (with a few tweaks)

$$\begin{cases} y^2 = x(x - A^2/B^2)(x - B^2/A^2), \\ \sqrt{\frac{A^2+B^2}{A^2-B^2}} \in k, \end{cases} \rightsquigarrow y^2 = x(x - a/b)(x - b/a).$$

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Classification

Some classification results

We present some classification results of models of Kummer lines[6] associated to an elliptic curve via:

- Galois representations and modular curves;
- the theta group.

We also give an application to isogeny volcanoes.

[6] R. Barbulescu, D. Robert and N. S. *Models of Kummer lines and Galois representations.* Cryptology ePrint Archive, Paper 2025/543. 2025.

Galois representations

Let E be an elliptic curve, set $E[N] = \langle P, Q \rangle$. If $\sigma \in \mathrm{Gal}(\overline{k}/k)$, then $\sigma(P), \sigma(Q) \in E[N]$, therefore

$$\sigma(P) = aP + cQ, \ \sigma(Q) = bP + dQ.$$

We have a Galois representation $\rho_{E,N} : \sigma \mapsto \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{GL}_2(\mathbb{Z}/N\mathbb{Z})$.

The image of $\rho_{E,N}$ gives extra information on the N-torsion of E. For instance, $\operatorname{im} \rho_{E,N} = \{\mathrm{id}\} \implies E[N] = E[N](k)$.

Some examples of modular curves (over $\mathbb{C}$)

$$\Gamma(N) = \{\gamma \in \mathrm{SL}_2(\mathbb{Z}) \mid \gamma \equiv \left(\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}\right) \bmod N\} = \ker(\mathrm{SL}_2(\mathbb{Z}) \twoheadrightarrow \mathrm{SL}_2(\mathbb{Z}/N\mathbb{Z}))$$

$\Gamma \subseteq \mathrm{SL}_2(\mathbb{Z})$ is a congruence subgroup if $\Gamma(N) \subseteq \Gamma$ for some $N \geq 1$.

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$\Gamma \subseteq \mathrm{SL}_2(\mathbb{Z})$ is a congruence subgroup if $\Gamma(N) \subseteq \Gamma$ for some $N \geq 1$.
$\mathrm{SL}_2(\mathbb{Z})$ acts on $\mathbb{H}^* = \mathbb{H} \cup \mathbb{Q} \cup \{\infty\}$ via

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \cdot \tau = \frac{a\tau + b}{c\tau + d}, \quad \begin{pmatrix} a & b \\ c & d \end{pmatrix} \cdot \frac{p}{q} = \frac{ap + bq}{cp + dq}, \quad \begin{pmatrix} a & b \\ c & d \end{pmatrix} \cdot \infty = \frac{a}{c}.$$

Set $Y(\Gamma) = \Gamma \backslash \mathbb{H}$ and $X(\Gamma) = \Gamma \backslash \mathbb{H}^* = Y(\Gamma) \cup \Gamma \backslash (\mathbb{Q} \cup \{\infty\})$.

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Set $Y(\Gamma) = \Gamma \backslash \mathbb{H}$ and $X(\Gamma) = \Gamma \backslash \mathbb{H}^* = Y(\Gamma) \cup \Gamma \backslash (\mathbb{Q} \cup \{\infty\})$.

- $X(N) = X(\Gamma(N))$;
- $X_0(N) = X(\Gamma)$ where $\Gamma = \{\gamma \in \mathrm{SL}_2(\mathbb{Z}) \mid \gamma \equiv \left(\begin{smallmatrix} * & * \\ 0 & * \end{smallmatrix}\right) \bmod N\}$;
- $X_0(M, MN) = X(\Gamma)$ where
  $\Gamma = \{\gamma \in \mathrm{SL}_2(\mathbb{Z}) \mid \gamma \equiv \left(\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}\right) \bmod M, \gamma \equiv \left(\begin{smallmatrix} * & * \\ 0 & * \end{smallmatrix}\right) \bmod MN\}$.

Shimura theory

Modular curves

Curve equation and rational points

Galois representations

Extra torsion information

Shimura theory

Shimura theory

Modular curves — Galois representations

Curve equation and rational points

Extra torsion information

E is an elliptic curve over $\mathbb{Q}$ with j-invariant $j_E \neq 0, 1728$.

Modular interpretation

Let G be a subgroup of $\mathrm{GL}_2(\mathbb{Z}/N\mathbb{Z})$ with $-\mathrm{id} \in G$ and $\det(G) = (\mathbb{Z}/N\mathbb{Z})^*$.
$\mathrm{im}\,\rho_{E,N}$ is conjugate in $\mathrm{GL}_2(\mathbb{Z}/N\mathbb{Z})$ to a subgroup of G if and only if $j_E \in p_G(X(G)(\mathbb{Q}))$, where $p_G : X(G) \twoheadrightarrow \mathbb{P}^1 \simeq X(1)$.

Application to models of Kummer lines

Proposition

the Kummer line of E admits a Montgomery model $M(a : b)$;
iff E has a rational subgroup $K \simeq \mathbb{Z}/4\mathbb{Z}$;
iff E admits a cyclic rational 4-isogeny;
iff $j_E \in p(X_0(4)(\mathbb{Q}))$ where $p : X_0(4) \to \mathbb{P}^1 \simeq X(1)$.

Proposition

the Kummer line of E admits a theta squared model $\theta_s(a : b)$;
iff the 2-torsion of E is rational and E has a rational subgroup $K \simeq \mathbb{Z}/4\mathbb{Z}$;
iff the 2-torsion of E is rational and E admits a cyclic rational 4-isogeny;
iff $j_E \in p(X_0(2,4)(\mathbb{Q}))$ where $p : X_0(2,4) \to \mathbb{P}^1 \simeq X(1)$.

Using the theta group

Consider $T \in E[2](k)$, we have seen that $t_T : \mathbb{P}^1 \to \mathbb{P}^1$ is well-defined and is a homography. Consider $M_T \in \mathrm{GL}_2(k)$ an affine lift of t_T, then $M_T^2 = \lambda_T \,\mathrm{id}$.

Type of a point

λ_T is unique up to a square, the class in $k^*/k^{*,2}$ is called the type of T.

On a finite field k, we have two types: squares (Montgomery type) and non-squares (anti-Montgomery type).

Some results related to the type

Theorem (Montgomery type)

The following assertions are equivalent:

1. the type of T is a square;
2. E has a Montgomery equation where $x(T) = 0$;
3. the Kummer line of E admits a model with ramification points

$$0_E = (1:0)^*, \quad T = (0:1), \quad T_2 = (\alpha:1), \quad T_3 = (1:\alpha) \qquad (\alpha \in \overline{k});$$

4. E has a rational cyclic subgroup K of degree 4 containing T;
5. there is a 4-torsion point T' above T which is rational on the Kummer line, i.e. $x(T') \in k$;...

The Kummer line of E admits a rational theta model $\theta(a:b)$ if and only if there are two rational points $T_1, T_2 \in E[2](k)$ both of Montgomery type.

Application to isogeny volcanoes

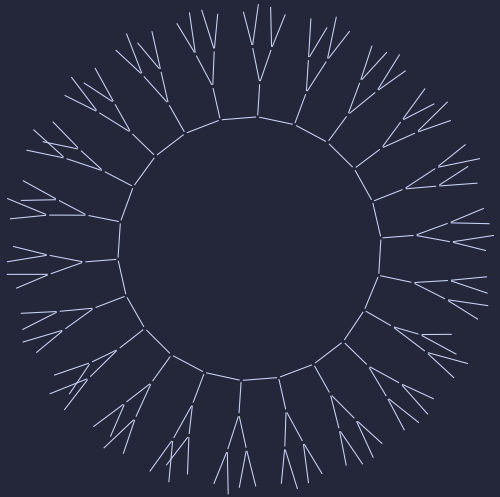


Figure: A 2-isogeny volcano

- Vertices: j-invariant of elliptic curves $E/\mathbb{F}_q$;
- Edges: 2-isogenies over $\mathbb{F}_q$.

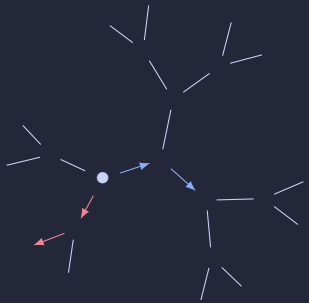
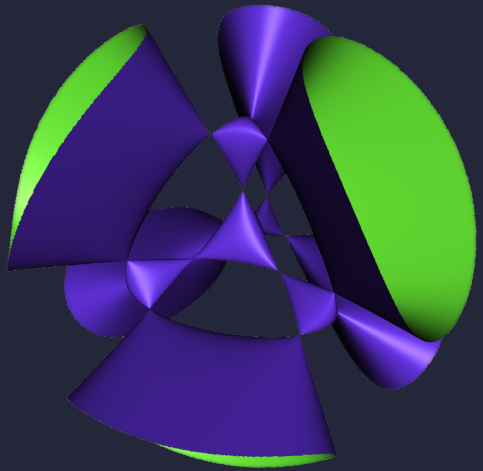


Figure: Finding 4-isogenies

Conclusion

The theta group is a great tool to study elliptic curves and abelian varieties more generally.

Thank you!